TempTrek Environmental Monitoring System

Version 2.5.5 — Official Installation & Deployment Guide

Document Type:	Installation & Administration Manual	**Target Platform:**	Windows 10 / 11 / Server (x64)
App Version:	v2.5.5	**Security Standard:**	DPAPI / AES-GCM / 21 CFR Part 11
GAMP 5 Category:	Category 4 (Configured) / Category 5	**Privilege Level:**	Windows Administrator (UAC Required)

1. System Requirements & Prerequisites

Before installing TempTrek Environmental Monitoring System, verify that the host machine meets the following minimum requirements:

Component	Minimum Specification	Recommended Specification
Operating System	Windows 10 x64 (Build 19041+)	Windows 11 x64 / Windows Server 2022
Architecture	x86-64 (64-bit)	x86-64 (64-bit)
Memory (RAM)	2 GB RAM	4 GB RAM or higher
Disk Space	250 MB available storage	2 GB+ (for multi-year CSV logs & backups)
User Privileges	Local Administrator rights (UAC)	Local Administrator rights (UAC)
Network Interface	Ethernet/Wi-Fi with LAN access	Dedicated Gigabit LAN for Modbus TCP
Dependencies	Bundled in MSI/EXE (No external runtime required)	Python 3.12 (Only if compiling from source)

2. Distribution Packages

TempTrek is distributed in two primary Windows deployment formats located in the dist/ directory:

- **Native Windows Installer (TempTrek-2.5.5-x64.msi):** Per-machine installation package. Installs to 64-bit Program Files, creates Start Menu and Desktop shortcuts, and supports standard Windows Apps uninstall & upgrade management.
- **Standalone Executable (TempTrek Environmental Monitoring System.exe):** Portable executable containing embedded Python runtime and all security/Modbus dependencies. Requires no installation step.
- **Inno Setup Executable (TempTrek Environmental Monitoring System_Setup.exe):** Alternative setup wizard compiled via installer.iss.

3. Step-by-Step Installation Procedure

Method A: Installing via Native Windows Installer (.MSI)

1. Locate **dist/TempTrek-2.5.5-x64.msi**.
2. Right-click the file and select **Install** or double-click to launch the Windows Installer setup wizard.
3. Accept the Windows User Account Control (UAC) prompt to allow administrative elevated access.
4. The installer automatically provisions binary files into C:\Program Files\TempTrek Environmental Monitoring System.
5. Shortcuts will automatically be created on the **Desktop** and in the **Start Menu** under *TempTrek*.

Method B: Deploying Standalone Executable (.EXE)

1. Copy **dist/TempTrek Environmental Monitoring System.exe** to your preferred application directory (e.g. C:\Program Files\TempTrek\).
2. Right-click the executable, select **Properties**, go to **Compatibility**, and check **Run this program as an administrator** (or accept the built-in PyInstaller manifest UAC elevation prompt).
3. Double-click the .exe to launch the system.

4. First-Time Launch & Security Setup

IMPORTANT SECURITY REQUIREMENT: TempTrek requires execution under a Windows Administrator account. Backup storage and directory ACL management rely on elevated SYSTEM privileges. Launch the application using *Run as Administrator*.

Initial Super Admin Setup

Upon launching the system for the first time on a machine:

1. The application detects that no active user database exists in %APPDATA%\TempTrek\config\users.json.
2. A **First-Time Super Admin Password Prompt** will appear.
3. Enter a master password fulfilling the security policy:
   - Minimum length: **6 characters**
   - Must be a new password different from existing credentials.
4. The password is hashed using **PBKDF2-SHA256** with unique cryptographic salting.

Application Storage & DPAPI Encryption

TempTrek stores application state and configuration securely across Windows directories:

Directory Path	Purpose & Security Rules
%APPDATA%\TempTrek\config	Stores devices.json, users.json, email_settings.json, and trend_settings.json. Protected via **Windows DPAPI** encryption.

%APPDATA%\TempTrek\live	Stores daily-rotated CSV logs ([IP]_[Port]_s[SlaveID]_[YYYY-MM-DD].csv) and alarm history.
%APPDATA%\TempTrek\reports	Destination for generated ReportLab PDF exports and approval audit records.
%ProgramData%\TempTrekBackups	Protected backup location. Access restricted exclusively to **Windows Administrators / SYSTEM** via strict ACLs. Holds password-encrypted .ttbackup archives.

5. Modbus TCP & Device Configuration

After initial login, configure target Modbus TCP devices under **Administration > Node Config**:

1. **Add Modbus Slave Node:** Enter Device IP, Port (default 502), Slave ID (e.g. 1), and **Time Interval to Store (sec)**.
2. **Configure Registers:** Click *Register Mapping* to map channels:
   - **Output Coil (FC01)** / **Input Coil (FC02)**: Boolean states (e.g. 0 = Normal, 1 = Alarm).
   - **Holding Register (FC03)** / **Input Register (FC04)**: Numeric parameters (FLOAT32, INT16, INT32, BCD).
3. **Decimal Precision:** Set scaling precision per channel to automatically format live cards and PDF trend graphs.
4. **Start Monitoring:** Click **START MONITORING** to initiate 1.0-second background polling threads.

6. Backup & Disaster Recovery

Navigate to **Administration > Backup & Restore** in an elevated Administrator session:

- **Backup Password Setup:** Click *SAVE SETTINGS* or *CREATE BACKUP NOW* to set a dedicated backup password.
- **Backup Encryption:** Backups produce .ttbackup zip archives encrypted with AES-GCM.
- **Restoration Safety:** Restoring an archive automatically freezes polling, validates checksums, creates a pre-restore safety snapshot, executes atomic transactional folder replacement with recovery journal rollback, and restarts the application.

7. Rebuilding Packages from Source

To rebuild distribution packages from Python source code:

1. Install Windows Python 3.12 (required for msilib support).
2. Install required build packages:

```
pip install pyinstaller cryptography>=46,<51 reportlab Pillow requests pymodbus certifi
```

3. Execute the automated master build script from project root:

```
build.bat
```

This generates both dist/TempTrek Environmental Monitoring System.exe and dist/TempTrek-2.5.5-x64.msi.

8. Verification & Test Suite Execution

To verify application integrity and security controls on a host system, run the automated test suite:

```
python -m unittest discover -s tempterk -p "test_*.py" -v
```

All 50 unit tests must pass, confirming DPAPI security, salted PBKDF2 hashing, atomic write protection, ACL generation, and restore rollback mechanics.

9. GAMP 5 Software Classification & GxP Compliance

Under the ISPE GAMP 5 framework (Good Automated Manufacturing Practice, Second Edition) for pharmaceutical and regulated GxP environments:

GAMP Category	Deployment Model & Classification Justification
Category 4 (Configured Software)	**Standard Production Deployment:** TempTrek is delivered as a pre-compiled software package (MSI/EXE). Site administrators configure Modbus device nodes, register mappings (FC01-FC04), threshold alarm rules, and user roles via standard UI forms without modifying source code.
Category 5 (Custom / Bespoke Software)	**Custom / Source Code Modification:** Applies if the underlying Python codebase (2.py, storage_security.py) is modified, extended, or customized for site-specific custom hardware protocols.

FDA 21 CFR Part 11 & EU Annex 11 Compliance Summary:

- **Audit Trail**: Comprehensive, chained SHA-256 audit logging of user actions, configuration changes, and viewed data.
- **Security & Integrity**: Windows DPAPI configuration encryption, PBKDF2 password hashing, and restricted administrator-only backup storage ACLs.
- **Report Signatures**: Password-authenticated data approvals; unapproved historical logs render reports with a *DRAFT* watermark.